

EFE DEMIRER

Electrical & Electronics Engineering Student

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SUMMARY

Senior Electrical and Electronics Engineering student at Yaşar University with hands-on laboratory and field engineering experience across RF & antenna test automation, Cartesian motion control systems, hardware interfacing, and solar telemetry. At ASELSAN REHİS Test Centers Directorate, developed automated C# GUI control platforms for planar Cartesian X-Y antenna scanners with GRBL G-Code and fail-safe limit switch interlocks, engineered Ethernet-based Real-Time Spectrum Analyzer API integrations featuring custom peak search algorithms, and conducted hands-on VNA calibrations and P1dB amplifier compression testing. At OPA Mühendislik, designed solar PV electrical infrastructures and co-engineered the SolarOPA autonomous AI monitoring and telemetry system.

EDUCATION

Yaşar University – İzmir

09/2022 – Present

B.Sc. in Electrical and Electronics Engineering | 100% Merit Scholarship | GPA: 3.73 / 4.00 (High Honor Standing)

Cem Bakioglu Anatolian High School – İzmir

09/2018 – 06/2022

High School Education | Salutatorian (Ranked 2nd in School) | Graduation Grade: 96.89 / 100

WORK EXPERIENCE

ASELSAN REHİS Test Centers Directorate

06/2026 – 08/2026

Intern Engineer

- Contributed to the development of Cartesian X-Y scanning automation and C# .NET graphical user interfaces (GUI) for planar near-field antenna test systems.
- Engineered API-based instrument control, Ethernet communication integration, and automated test execution scripts.
- Designed a real-time signal peak detection algorithm with dynamic thresholding for automated spectral measurements.
- Performed Vector Network Analyzer (VNA) calibrations, port reference plane extensions to cable ends, RF component characterization, and power amplifier testing.
- Conducted P1dB compression tests to measure amplifier saturation and non-linear compression characteristics.
- Performed near-field chamber measurements and analyzed RF radar-absorbent material performance using the NRL Arch Test method.
- Evaluated spectrum analyzer specifications and conducted laboratory/field spectral signal monitoring.
- Supported the planning of RF test scenarios and the execution of automated measurement test suites.
- Developed hardware safety interlocks with Limit Switches, stepper motor drive systems, and GRBL G-Code integration on CNC hardware shields.

OPA Mühendislik

07/2025 – 08/2025 & 08/2026 – 09/2026

Intern Engineer

- Performed ground resistance measurements, residual current leakage tests, and technical compliance reporting.
- Participated in field installation, electrical panel assembly, and CAD engineering for solar PV and smart grid/storage infrastructures.
- Prepared solar PV electrical projects and single-line schematics using AutoCAD.
- Executed solar plant simulations, yield analyses, and production performance forecasting using PVsyst.
- Designed power factor compensation panels, calculated capacitor bank sizing, and configured reactive power control relays.
- Calculated power loads, selected cable cross-sections, and supported installations for electric vehicle (EV) charging stations.
- Assembled electrical distribution boards, connecting circuit breakers and residual current devices (RCD) for PV arrays.
- Participated in periodic inspections of transformer substations under High Voltage Facility Operation Responsibility regulations.

TECHNICAL SKILLS

- Programming & Software Development:** C#, .NET, Windows Forms, Graphical User Interfaces (GUI), Python, Arduino IDE
- Test Automation & Instrument Control:** Test Automation, Hardware Interfacing, API Integration, SCPI, Serial/Ethernet Communication
- RF & Antenna Testing:** Vector Network Analyzer (VNA), Spectrum Analyzer, RF Component Characterization, Near-Field Chamber Scanning
- RF Measurement & Characterization:** RF Power Amplifier Measurements, P1dB Compression Testing, RF Absorber NRL Arch Testing
- Motion & Scanning Systems:** Planar Cartesian X-Y Motion Control, Stepper Motor Drives, GRBL G-Code Firmware, Limit Switch Interlocks
- Electrical & Power Systems:** AutoCAD, PVsyst, ETAP, Solar PV Systems, Single-Line Diagrams, Power Factor Correction, Capacitor Sizing
- Simulation & Modeling:** MATLAB & Simulink, LTspice, Proteus

LANGUAGES & PROFESSIONAL COMPETENCIES

- Languages:** English B2 (Upper-Intermediate), Yaşar University School of Foreign Languages certified
- Professional Competencies:** Cross-functional Team Collaboration, Technical Documentation & Reporting, Analytical Problem Solving

ADDITIONAL INFORMATION

- Personal Details:** Driving License: Class B | Military Service: Postponed until 31/12/2031
- Interests:** Electronics Repair (DIY), Football, Camping, Swimming, Travel, Agricultural Production

REFERENCES

Prof. Dr. Mustafa SEÇMEN – Dean, Faculty of Engineering / Yaşar University | mustafa.secmen@yasar.edu.tr | +90 232 570 8232

Dr. Ali Haluk NALBANTOĞLU – Faculty Member / Yaşar University, Former Director of Electronic Warfare at ASELSAN | haluk.nalbantoglu@yasar.edu.tr | +90 533 236 1399

Hüseyin Talha ÇULHA – RF Test Engineer / ASELSAN REHİS | htalhaulha@aselsan.com | +90 531 950 4360

Murat YALÇINKAYA – Systems Engineer / ASELSAN REHİS via EHSİM | yalcinkayam@gmail.com | +90 538 847 9001

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